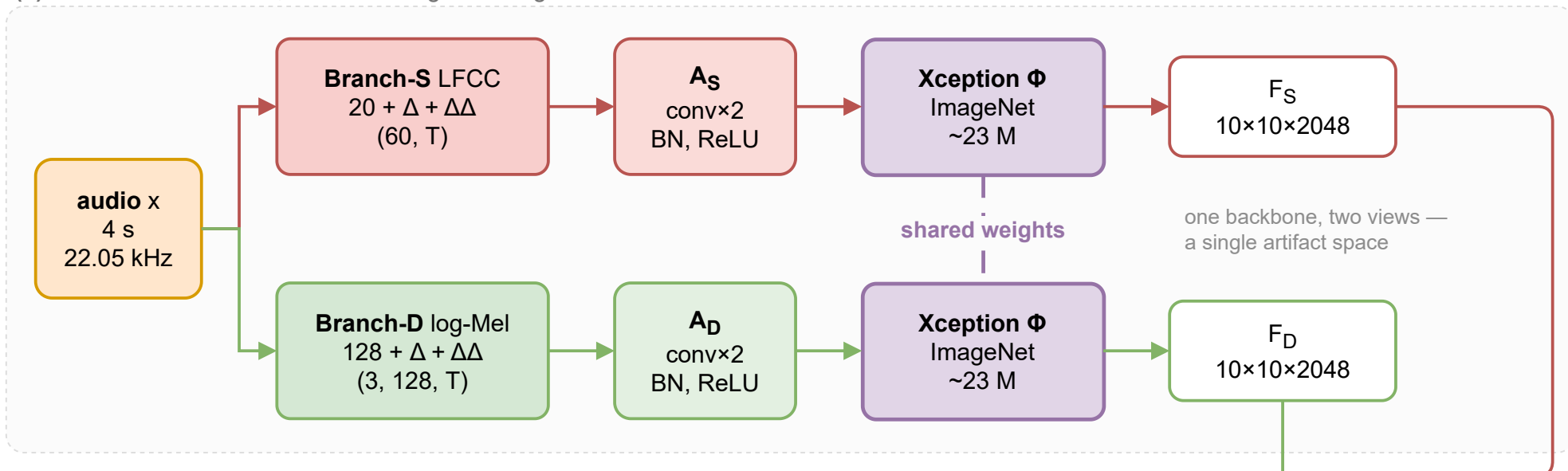


(a) dual-branch encoder, Siamese weight sharing



(b) cross-attention spectral fusion and the three objective heads

